

Final-year Computer Science (AI/ML) student at VIT-AP. I build systems end-to-end, from a multi-algorithm movie recommender to an RL visualizer, and this summer ran signal-processing analyses on intracranial neural recordings through Neuromatch Academy's computational neuroscience program.

SKILLS

PROGRAMMING	JAVA	PYTHON		SQL			
	ML	NLP	RL				
AI							
TOOLS	GIT	REST APIS		MNE-PYTHON	FIGMA	CANVA	

PROJECTS

RecommenToMe Movie Recommender System

PYTHON · FLASK · SCIKIT-LEARN · JS

- Implemented 5 recommendation algorithms from scratch (item-based CF, TF-IDF content-based, SGD matrix factorization, hybrid, Bayesian popularity) behind a shared interface, served through 12 Flask REST endpoints
- Built an evaluation harness on MovieLens with a leakage-free per-user temporal split, scoring RMSE, Precision@10, recall, coverage, diversity, and novelty; tuned the hybrid's blend weight with a parameter sweep
- Vanilla JS frontend compares every algorithm's picks for any movie side by side; verified personalization quantitatively (cross-user top-10 overlap of 0.52 for popularity vs 0.03 for item CF)
- **IN PROGRESS** automated pytest suite over the Flask endpoints (schema, input validation, edge cases), run on each push via GitHub Actions

learn-to-RL Interactive RL Visualizer

VANILLA JS · CANVAS

- Browser-based playground visualizing RL algorithms (Q-learning live; SARSA, DQN, policy gradient planned) with zero install and zero dependencies
- Live Q-table as directional arrows on a 7×7 grid with real-time reward charts and ϵ -decay tracking; full Q-learning update rule in ~200 lines of vanilla JS, with an interactive code walkthrough

StudEx Full-Stack Web Application

NODE · EXPRESS · POSTGRESQL

- Roll-number lookup returning a student's per-subject marks alongside a rule-based remark derived from those scores; unrecognised roll numbers return a recheck prompt rather than an error

RESEARCH EXPERIENCE

Neuromatch Academy Computational Neuroscience · Project Pod

JUL 2026

- A summer computational neuroscience program giving students the opportunity to research open datasets; led a four-student project pod analyzing intracranial (ECoG) recordings from an n-back working-memory task, testing whether broadband high-gamma power (70–170 Hz) tracks memory load
- Built the team's MNE + FOOOF preprocessing pipeline as primary coder
- Traced an initial null result to a normalization artifact; after correcting it, found a load effect in 2 of 3 subjects, above-chance multiclass load decoding in all 3 (permutation $p < .001$), and a load-dependent shift in the aperiodic (1/f) spectral exponent ($p < .002$)
- Presented findings to course TAs; prepared results for abstract submission

LEADERSHIP & EXPERIENCE

Vice President Open Source Community · VIT-AP

OCT 2024 – MAY 2026

- Led strategic direction for an 85+ member community across technical and creative tracks; organized 10+ workshops and hackathons, after starting as Events / Track Lead

President Madhya Bharat Association · VIT-AP

MAY 2024 – MAY 2026

- Managed cross-functional teams of 50+ across design, content, and logistics; improved member retention 30% and event attendance 60%

Design Team Lead Nrityam, Anchoring Club, GDG on Campus · VIT-AP

MAY 2024 – MAY 2025

- Led design teams concurrently across 3 student organizations, delivering visual assets for 5+ events with audiences of 500+

EDUCATION

B.Tech, Computer Science Engineering VIT-AP University

2023 – 2027

- CGPA: 8.5 / 10 · Coursework: Machine Learning, Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Reinforcement Learning
- Class XII: 93% · Class X: 93%

CERTIFICATES & AWARDS

Kaggle Contest: 3rd Place · Udemy: Full Stack Web Development Bootcamp · NPTEL: Human Computer Interaction (Elite, Gold) · NPTEL: Classics in Neuroscience (Elite, Silver) · AI Foundations: Thinking Machines

LANGUAGES

ENGLISH: FLUENT	HINDI: FLUENT	KOREAN: INTERMEDIATE	GERMAN: ELEMENTARY
-----------------	---------------	----------------------	--------------------